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RFID-ENABLED MEDICAL INJECTORS AND ARTIFICIAL INTELLIGENCE (AI) PLATFORM FOR PRE-INJECTION INTERROGATION

Abstract

In some embodiments, an AI agent may be operable to conduct a pre-injection interview with a patient (e.g., as part of a self-injection process) prior to authorizing an injection for the patient. The interrogation may be designed to identify whether the patient is at an increased risk of adverse side effects, based on historical data accessible to the AI agent. In accordance with some embodiments, the AI agent may be programmed to learn and improve the questions to ask patient (e.g., for a given fluid agent, in particular circumstances, for specific types of questions, or otherwise), based on inputs received from patients who have previously received injections and experienced adverse side effects. In some embodiments, the AI agent may further be programmed to implement one or more mitigation measures based on a result of such an interrogation.

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Background/Summary

CLAIM OF PRIORITY [0001] The present application is a Continuation Application of PCT Patent Application No. PCT/IB2023/000651, filed on Nov. 6, 2023 and titled RFID-ENABLED MEDICAL INJECTORS AND ARTIFICIAL INTELLIGENCE (AI) PLATFORM FOR PRE-INJECTION INTERROGATION, which PCT Application claims the benefit of U.S. Provisional Application No. 63/423,041 filed on Nov. 6, 2022 and titled RFID-ENABLED MEDICAL INJECTORS AND AI PLATFORM FOR PRE-INJECTION INTERROGATION. The entirety of each of these Applications is incorporated by reference herein for all purposes.

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BACKGROUND

[0003] Every year, staggering numbers of people become infected and die from a variety of diseases, some of which are preventable (or the severity of which could have been mitigated) through injectable medicines (i.e., medicines delivered via use of a syringe or other needle-type medical delivery device). Although injectable medicines have led to a dramatic decline in the number of cases of several infectious diseases (or the severity of symptoms or instances of death resulting therefrom), some of these diseases remain quite common and new ones are emerging. In many instances, large populations of the world, particularly in developing or economically disadvantaged countries, suffer from the spread of preventable diseases due to ineffective injectable medicine/vaccine delivery programs, either because of poor implementation, lack of available medicines/vaccines, lack of sufficient skilled persons to administer injections, poor tracking and management of where injections have been administered vs. where there is still a need for administration, or combinations thereof. The 19th Century Pandemic, have further highlighted the inadequacies of existing vaccine and other injectable medicine administration programs. This Pandemic is an example of a situation in which a populace would benefit from (i) being able to verifiably and efficiently track who has been vaccinated (as well as perhaps tracking where, when and with what particular vaccine); (ii) allowing persons to efficiently provide evidence of having been vaccinated (and, in some embodiments, a current vaccination status, such as for circumstances in which vaccine effectiveness requires booster shots or additional injections), (iii) identifying areas of the population that have ongoing vaccination needs; and/or (iv) enabling a wider-ranging vaccination effort by providing means for less skilled persons to administer vaccinations.

[0004] Additionally, Applicant has recognized that there is an ever-growing need for patients to self-administer medicines (whether by injection or otherwise), and that such self-administration events may involve various fact patterns in which the patient would benefit from an interaction or interview with a health care professional or other relevant expert. For example, a patient may be experiencing certain symptoms (e.g., a fever, fatigue, updates in medical history), or have

questions, that were not present when the patient was initially receiving a prescription for the medicine. In another example, there may be updates in the status of the medicine (generally or for a particular batch or dose), based on newly acquired data, and which should ideally be communicated to the patient prior to the next dose of medicine being administered.

[0005] Unfortunately, in many instances there is a lack of sufficient medical personnel to assist such patients with any last-minute questions or concerns at the point of self-administration, to interview a patient prior to self-administration and/or to communicate updated data about the medicine to the patient. Thus, there is a risk that patients may be self-administering medicines inappropriately (e.g., when they shouldn't be, due to then-present physical conditions or updated medical history that may adversely interact with the medicine and/or due to inadequate up-to-date information about the medicine). Additionally, there is an increased risk that patient may be foregoing a self-administration event due to concerns or questions that they are not able to adequately address in a timely manner.

[0006] Existing systems and infrastructure have proven inadequate to satisfy the foregoing needs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] An understanding of embodiments described herein and many of the attendant advantages thereof may be readily obtained by reference to the following detailed description when considered with the accompanying drawings, wherein:

[0008] FIG. 1 is an example block diagram of a system consistent with at least some embodiments described herein;

[0009] FIG. 2 is an example block diagram of a server device consistent with at least some embodiments described herein; and

[0010] FIG. 3 is an example process flow diagram of an example process consistent with at least some embodiments described herein.

DETAILED DESCRIPTION

[0011] Embodiments described herein relate to injectors for administering single doses of fluid agents that are equipped with a data storage mechanism that allows for electronically recognizing, tracking and/or authorizing the administration of the single doses into patient. In some embodiments, an electronic platform is operable to facilitate (i) transmission of data from an injector to an injection provider device and then to the electronic processing system of the platform, to facilitate the authorization and tracking of the administration of a dose of the fluid agent from the injector into a patient; and/or (ii) exchange of data with a patient's device that allows the patient to receive and store a digital receipt of having had the single dose of the fluid agent administered from the injector. The electronic platform further allows for healthcare systems and authorities to generate and utilize an information layer associated with the injectors that are equipped with the data mechanism.

[0012] In accordance with some embodiments, such a data layer may comprise data comprising one or more adverse side effects or increased risks that some prior patients have experienced, that correspond to one or more specific circumstances, conditions or characteristics of the patients at the time of injection. The data layer and/or AI-driven platform accessing such a data layer may be operable to determine the correlation and determine that, if the one or more specific circumstances, conditions or characteristics of the patient had been recognized at the time of injection (and the correlation of these to the one or more adverse side effects or increased risks had been known), one or more measures or actions may have been possible to implement in order to mitigated or avoided via one or more mitigation measures ("mitigation measures"). Such mitigation measures may subsequently be put in place for future patients that are identified as possessing or having the

relevant one or more specific circumstances, conditions or characteristics. For example, if data from prior patients indicates that patients with a particular condition or characteristic (e.g., patients prone to increased heart rate or patients with high blood pressure) are more likely to experience negative side effects upon being injected with a particular drug or medicament (or dosage thereof), an AI-driven platform may be programmed to not only recognize the correlation but also to update pre-injection protocols designed to identify patients with such circumstances, conditions or characteristics prior to authorizing an injection of said drug or medicament (or particular dosage thereof). For example, the AI-driven platform may update or modify a set of questions or interrogation protocol to be presented to a patient prior to an injection of the drug or medicament being authorized, in a manner designed to identify whether the patient's profile includes the one or more circumstances, conditions or characteristics (e.g., add a question regarding the patient's heart health, prompt the patient to enter a current blood pressures, etc.). Further, the AI-driven platform may update or modify an injection authorization protocol for patients who are identified as having the one or more circumstances, conditions or characteristics (e.g., by putting the patient in touch with a health care professional, instructing the patient to inject a lower dosage of the drug or medicament, or not authorize the injection until a health care professional discusses the additional risks or the anticipated increased likelihood of adverse side effects with the patient).

[0013] An injector, as the term is used herein, refers to a device that is filled with a single dose of a fluid agent, such as a vaccine or medicament (which may, in some embodiments, include a lyophilized component that may be reconstituted into a fluid prior to injection). An injector may be made of plastic, glass or another material and the embodiments described herein are not dependent on an injector being made from any particular material. An injector may comprise, for example, an auto-injector that is spring loaded or has another mechanism for assisting with the injection of the fluid agent, or a manual injector that relies on the user to operate the injection mechanism.

Examples of manual injectors include plunger-type syringes or plastic vial injectors (such as blow-fill-seal ("BFS") injectors) that requires a pressure or squeezing action on the part of the injection provider (e.g., a nurse or the patient, in the case of self-administered injections) in order to squeeze or otherwise force the fluid agent out of the vial or other component comprising the injector for containing the fluid agent. Examples of an injector device that may be utilized in embodiments described herein can be found in PCT Application No. PCT/US21/25683 filed on Apr. 3, 2021 in the name of Koska et al. and entitled SYSTEMS AND METHODS FOR PRE-FILLED MEDICAL DELIVERY DEVICES and in PCT Application No. PCT/US18/61696, filed on Nov. 16, 2018 in the name of Koska et al. and entitled SYSTEMS AND METHODS FOR FLUID DELIVERY MANIFOLDS, each of which PCT applications are incorporated by reference herein for all purposes.

[0014] It should be noted that injectors that are pre-filled with single doses of a fluid agent may be pre-filled during a manufacturing process (pre-filled injectors), such as the BFS injectors described in the afore-referenced PCT applications to Koska et al., or filled by injection provider, such as by extracting a single dose from a multi-dose container (field-fillable or field-filled injectors).

Injectors that are field-filled or field-fillable are ones in which an injection provider extracts a single dose of the liquid agent from another container (e.g., a multi-dose glass vial) into the injector prior to administering the injection. In the field-filled injectors, a dose of the liquid agent administered via the injector is not unique to the injector and thus an injection provider may need to provide two pieces of data to identify both the injector and an identifier of the multi-dose vial from which the dose was extracted. In a pre-filled injector, a unique identifier of the injector inherently can identify one or both of the injector itself and the dose of the fluid agent contained therein. Identifying a dose of fluid agent may be desirable, for example, in embodiments in which information such as the batch number, type of fluid agent, manufacturer, date of manufacture of the dose, etc. are of interest. The embodiments described herein are primarily focused on pre-filled injectors.

[0015] In accordance with embodiments described herein, injectors are equipped with data storage mechanisms that uniquely identify each injector, such as by storing a unique identifier for each injector that is generated at the time the injector is manufactured (and, in the case of pre-filled injectors, at the time the injector is pre-filled with a dose of fluid agent). Such data storage mechanisms may comprise, for example, a Near Field Communication (NFC) chip, a Radio Frequency Identification (RFID) chip, a QR code, a bar code or any other machine-readable mechanism that allows for a unique identifier of an injector to be stored in association with the injector (e.g., attached to, or embedded in, the injector or a portion thereof, such as on or underneath a label on a label-portion of the injector).

[0016] It should be noted that in some embodiments, the unique identifier and/or other data stored on an NFC chip or other data storage mechanism may be stored in encrypted form. It should further be noted that, in some embodiments, communications with the NFC chip (or other data storage mechanism) and/or the Injection Management App may be done in an encrypted form (e.g., using public key-private key protocols) or utilizing other security measures (e.g., dual factor authentication or blockchain technology).

[0017] In accordance with some embodiments, an Injection Management System may comprise providing an Injection Management App for facilitating the verification/authorization of injections via injectors equipped with data storage mechanisms as described herein. An Injection Management App may be utilized by individuals to communicate with a centralized or decentralized (e.g., blockchain-based) system for facilitating the creation and utilization of an information layer or electronic record keeping based on use of the injectors. In accordance with some embodiments, the Injection Management App may include an injection provider portal or platform, via which injection providers or other individuals who administer injections of fluid agents using injectors as described herein (“injection providers”) may sign in to the system and obtain access to functionality for such injection providers. For example, an injection provider may utilize the Injection Management App to confirm that a particular dose in a pre-filled injectors is valid and authorized for use (e.g., not counterfeit, not expired, and not subject to a recall), prior to administering it to a patient, and to record with the Injection Management System each injection event comprising an instance of the injection provider having provided a dose of a fluid agent packaged in a uniquely identified injector to a particular patient. Similarly, the Injection Management App may include a patient portal or platform, via which individuals receiving injections (“patients”) may sign in to the system to access functionality for patients (e.g., obtaining a digital receipt, to be stored on a mobile device of the patient, that can be shared in some form with third parties to offer verification of having been vaccinated with a particular vaccine or otherwise having received a dose of a particular fluid agent).

[0018] In some embodiments and circumstances, a given individual may serve as both an injection provider and a patient (“self-injection” embodiments). In some embodiments, a given individual may serve as both an injection provider and a patient for a given injection event (e.g., the individual can self-administer an injection). In self-injection and other embodiments, an Injection Management App may be operable to conduct a pre-injection interview or interrogation (e.g., of the patient directly, with the patient using the Injection Management App as part of the self-injection process) or may facilitate an injection provider to conduct such a pre-injection interview or interrogation of the patient (e.g., by prompting the injection provider to ask specific questions to the patient and enter the patient's answers). In accordance with some embodiments, such a pre-injection interview or interrogation (referred to as an interrogation herein) may consist of several questions that are driven by an Artificial Intelligence (AI) engine or algorithm (referred to as an AI agent here). The questions may be specific to the patient (e.g., based on the patient's medical history or other patient data) and/or fluid agent specific (e.g., based on the unique identifier of the pre-filled injector and thus based on the specific fluid agent, batch number and/or dose, as corresponding to the unique identifier). In accordance with some embodiments, the AI agent may

be programmed to learn and improve the questions to ask patient (e.g., for a given fluid agent, in particular circumstances, for specific types of questions, or otherwise). For example, over time the AI agent may learn to ask patients living in a particular geographic area during a particular time of year about specific symptoms they may be currently experiencing (e.g., patients living in South America during flu season may be asked about flu-like symptoms, prior to an injection of a particular fluid agent being authorized by the Injection Management App). In another example, the AI agent may learn, based on after-injection results or patient feedback that the AI agent may access, that asking whether the patient has recently eaten a meal prior to authorizing the patient to self-administer a dose of fluid agent X is important because patients who previously self-administered a dose of fluid agent X complained of nausea if they took it on an empty stomach.

[0019] Applicant notes that utilizing a pre-filled injector that is uniquely identifiable and providing an Injection Management App that is operable to conduct a pre-injection interrogation via an AI agent and/or recognize an injection event in which the injector is utilized offers one or more advantages over existing capabilities of healthcare systems. For example, the pre-injection interrogation may be helpful in reassuring a patient conducting a self-injection that the injection is safe to administer even if the patient's medical history or current medical condition is different from that when the injection was first prescribed by a doctor (e.g., the patient may feel better about administering the injection knowing that the App asked questions and received answers specific to the patient's current health conditions and still authorized the patient to self-administer the injection).

[0020] Embodiments presented herein are descriptive of systems, apparatus, interfaces, methods, and articles of manufacture for managing, tracking, authorizing and verifying injection events in which a patient receives an injection of a dose of fluid agent from an injector that is equipped with a unique identifier. Applicant describes herein various improvements that can be made to injectors such as BFS vials or a traditional glass vials by including within, or attaching to, the injector a unique identifier. The unique identifier may be in the form of a physical component such as an NFC or RFID chip, or a QR code. Irrespective of the form in which the unique identifier is embodied, the presence of the unique identifier as being associated with each injector allows for a layer of information to be created around, or associated with, the injector (or the dose contained within the injector, particularly in the case of pre-filled injectors) and/or a patient who receives an injection of a dose of the fluid agent packaged in the injector. Although various embodiments are described herein as being applicable to an injector comprising a BFS vial injector, it should be understood that at least some of these embodiments may also be applied to a glass or other vial or syringe system that can benefit from the generation of data and new functionalities described herein. Thus, it should be understood that when reference is made to a BFS vial or other particular type of injector, that references it not intended to be limiting and in various embodiments the same or similar functionality may be provided with respect to other types of injectors that serve as containers for fluid agents. Similarly, although reference is made herein in some embodiments to an NFC-chip-enabled injector, such reference is intended to refer to any type of injector (whether it be a BFS vial mechanism, a traditional glass vial or otherwise) that is associated with a unique identifier that is readable via a software app as described herein (e.g., has an NFC chip or RFID chip embedded within it or attached to it in some manner, or a QR code attached to it, that allows a unique identifier of the NFC chip to uniquely identify the corresponding injector and the specific dose of fluid agent contained therein). An injector that has such a unique identifier corresponding to it is also referred to as a Unique ID injector herein. It should be noted that if the term "injector" is used rather than "Unique ID injector", it should not be interpreted to imply that a unique identifier is not associated with that injector in the context described, as the term "injector" is used as a shorthand for "Unique ID injector" in some passages. Further, it should be noted that a unique identifier or data storage mechanism comprising a unique identifier being attached to, embedded in, printed or embossed on or otherwise associated with an injector may comprise the unique identifier

or data storage mechanism storing the unique identifier with an injector may, in some embodiments, mean that the unique identifier or data storage mechanism is attached to or otherwise associated with a particular component (e.g., a BFS vial or a label portion of a BFS vial) or packaging of the injector (e.g., an NFC chip may be attached to foil packaging of an injector).

[0021] Applicant has recognized that including a data storage mechanism or a means of accessing a data storage mechanism via a unique identifier, such as via an NFC chip, with each injector would allow certain new and useful benefits to be realized, such as: (i) allowing the injector to connect to the Internet of Things (IoT) and/or communicate with remote servers and/or other devices connected to IoT; (ii) enabling an AI agent to conduct a pre-injection interrogation process with a patient or other person performing an injection; (iii) allowing a user (whether such user be an injection provider, a patient receiving an injection, or a parent, guardian or other person associated with the patient receiving the injection) of the Unique ID Injectors to use a specially programmed software app on a mobile device or other device (the app having been previously been downloaded onto the device), referred to as an Injection Management App herein, to input data or information into the App and/or to allow the App to read information from the data storage mechanism (and, in some embodiments, to write data to the data storage mechanism, such as if the data storage mechanism comprises an NFC or RFID chip); (iv) allowing for the verifying, authorizing, tracking, confirming and rewarding of administration of fluid agents from the injectors into patients; and (v) allowing for follow-up communications (e.g., text messages or messages output via the Injection Management App) to be output to patients who have received injections from such injectors, such as to remind the patients to self-inject additional doses or to return to a clinic for additional doses, ask about possible adverse reactions (e.g., to be utilized by the AI agent in future pre-injection interrogations), receive confirmations that additional verified or authorized doses have been injected into the patient as recommended, etc.

[0022] It should be noted that there may be at least three different types of users involved in the injection of a fluid agent from an NFC-chip-enabled injector: (i) an injection provider such as a healthcare worker; (ii) a patient into whom the fluid agent is injected; and (iii) a guardian (e.g., parent), friend or family member of a patient into whom the fluid agent is being injected (which, in some cases, may be the person in a village with whom the patient shares a mobile device; collectively such persons who may be associated with a patient are referred to as patient liaison herein). Not all of these types of users may be involved in a given injection. In accordance with some embodiments, one or more of these users may access a different platform, portal or version of an Injection Management App (or access different aspects, functionalities or pages of such an App) on their mobile device to facilitate one or more of the embodiments described herein. For example, an injection provider may access and utilize a first version or aspect of an Injection Management App (referred to as a Provider Platform of the Injection Management App) while a patient or patient liaison may access or utilize a second version or aspect of the Injection Management App (referred to as a Patient Platform of the Injection Management App).

[0023] In some embodiments, the Patient Platform may facilitate the tracking of the vaccines or other fluid agents administered to a particular patient and provide for an AI agent to conduct a pre-injection interrogation with the patient, healthcare provider or parent or guardian of the patient. It should be noted that, for purposes of brevity, the pre-injection interrogation process is primarily described herein as being conducted with the patient. It should be understood, however, that such a process (or a substantially similar process) may be conducted with a healthcare provider, parent or guardian of the patient (e.g., these individuals may answer the questions posed via such a pre-injection interrogation process, on behalf of the patient).

[0024] In accordance with some embodiments, the Injection Management App may allow for communication of data to/from a remote server of a service that provides or manages the Injection Management App and the injection data collected thereby (such service being referred to as an Injection Management Service herein). In some embodiments, the Injection Management Service

may be involved with (e.g., control or have a business relationship with an entity that so controls) the manufacturing and/or distribution of the Unique ID Injectors as described herein and store records of Unique ID Injectors that were manufactured in accordance with embodiments described herein and that are to be tracked from the point of manufacture to the point of injection, in accordance with at least some embodiments described herein. In some embodiments, the Injection Management Service may operate or control (directly or indirectly) the pre-injection interrogation process as described herein (e.g., as described with reference to process **300** of FIG. **3**) and/or the AI agent via which the pre-injection interrogation process is conducted.

[0025] In some embodiments, data collected via an Injection Management App may be cached and stored in local memory when the device on which the app is downloaded is not able to communicate with the Injection Management Service (e.g., when the device is not connected to Wi-Fi or does not have a sufficient cellular service connection) and then transfer the data to the Injection Management Service once a sufficient communication connection is established.

[0026] In accordance with some embodiments, the Injection Management Service may open a record in a database for each Unique ID Injector that is manufactured in accordance with embodiments described herein. In some embodiments, the Injection Management Service may store in that record the unique identifier of the NFC or RFID chip embedded in or attached to the vial in association with other information corresponding to the dose of fluid agent (e.g., time and place of manufacture; batch, lot and/or strip number; type of fluid agent; dosage of fluid agent; expiration date of fluid agent; etc.). Subsequently, when a user (e.g., an injection provider, patient or patient liaison) taps the Unique ID Injector to their mobile device, the Injection Management App may read the unique identifier stored in the data storage mechanism of the injector (e.g., the NFC chip identifier) and communicate with the Injection Management Service to retrieve information associated with the vial/chip in order to (i) verify or authorize the injection (e.g., check that the fluid agent is not expired or is not subject to a recall; conduct a pre-injection interrogation); (ii) open or create a new record (or update information in an existing data record) to indicate that the dose of fluid agent corresponding to that identifier is being injected, which may also include storing an indication of various new information obtained from the user or based on the tap of the vial to the phone (e.g., time/place of the injection; identifier of a injection provider, if any, providing the injection; identifier of the patient and/or an identifier (e.g., telephone number) of a mobile device associated with the patient), also referred to as an injection event record herein.

[0027] In some embodiments, the Injection Management Service operates an Injection Management System that comprises one or more servers operable to communicate with user mobile devices that have the Injection Management App installed thereon and are registered devices and/or users of the System. In some embodiments, the Injection Management System utilizes a blockchain type of distributed network of servers for storing the injection event records (e.g., for security purposes).

[0028] In some embodiments, in addition to or in lieu of contacting the Injection Management Service to verify or authorize the injection once a user taps the vial to a mobile device on which the Injection Management App is open, the Injection Management App may read information in addition to the chip identifier from the data storage mechanism (e.g., an NFC or RFID chip may store readable data such as expiration date, dosage, type of fluid agent contained in the vial) and use this information to authorize the injection and output to the user that he/she can move forward with the injection.

[0029] In accordance with some embodiments in which a Unique ID Injector comprises a BFS vial mechanism, Applicant envisions that the pre-filled BFS vial that is filled with a fluid agent during the BFS manufacturing process described herein may have embedded within its plastic, or otherwise attached to or included within it or on a surface or portion thereof, an NFC chip (the term “embedded” will be used for purposes of brevity but is intended to encompass all manner of attachment, impregnation, embedding or addition to, into or onto the vial, whether it be a BFS vial,

another type of plastic vial or a glass vial). In other embodiments, a data storage device (whether an NFC chip, an RFID chip, or other form) may be glued on or otherwise attached to an injector during the manufacturing process (e.g., it may be glued on or otherwise attached underneath a label, on a label portion of the BFS vial).

[0030] Turning now to FIG. 1, a block diagram of a system **100** according to some embodiments is shown. In some embodiments, the system **100** may comprise a plurality of node devices **102a-n**, a network **104**, a management device **106**, and/or a server device **110**. According to some embodiments, any or all of the devices **102a-n**, **106**, **110** may comprise and/or be in communication with a data storage and/or memory device **140-1a-n**, **140-2**. Each node device **102a-n** may comprise a local memory device **140-1a-n**, for example, and/or the server device **110** may comprise a network memory device **140-2**. As depicted in FIG. 1, any or all of the devices **102a-n**, **106**, **110**, **140-1a-n**, **140-2** (or any combinations thereof) may be in communication via the network **104**. In some embodiments, communications between and/or within the devices **102a-n**, **106**, **110**, **140-1a-n**, **140-2** of the system **100** may be utilized to provide and manage a distributed injection events transaction ledger. The server device **110** may, for example, interface with one or more of the node devices **102a-n** and/or the management device **106** to execute multiple instances of specially-programmed chain code (not depicted) stored in any or all of the memory devices **140-1a-n**, **140-2** and/or provide a specially-structured interface via which a user participating in an injection event transaction may obtain, verify, and/or modify status information pertaining to an injection event transaction.

[0031] Fewer or more components **102a-n**, **104**, **106**, **110**, **140-1a-n**, **140-2** and/or various configurations of the depicted components **102a-n**, **104**, **106**, **110**, **140-1a-n**, **140-2** may be included in the system **100** without deviating from the scope of embodiments described herein. In some embodiments, the components **102a-n**, **104**, **106**, **110**, **140-1a-n**, **140-2** may be similar in configuration and/or functionality to similarly named and/or numbered components as described herein. In some embodiments, the system **100** (and/or portion thereof) may comprise a distributed injection event management program, system, and/or platform programmed and/or otherwise configured to execute, conduct, and/or facilitate the methods described herein, including method **400**, and/or portions thereof.

[0032] The node devices **102a-n**, in some embodiments, may comprise any types or configurations of computing, mobile electronic, network, user, and/or communication devices that are or become known or practicable. The node devices **102a-n** may, for example, comprise one or more electronic devices, such as mobile devices comprising cellular and/or wireless telephones, such as an iPhone® (also manufactured by Apple®, Inc.) or an LG Optimus™ Zone™ 3 smart phone manufactured by LG® Electronics, Inc. of San Diego, CA, and running the Android® operating system from Google®, Inc. of Mountain View, CA. In some embodiments, the node devices **102a-n** may comprise devices owned and/or operated by one or more users, such as patients, injection providers or third parties seeking to confirm information about a particular injection event transaction. According to some embodiments, the node devices **102a-n** may communicate with the server device **110** via the network **104** to conduct injection verification inquiries and/or processes and/or to record, store, confirm or update information about an injection event (e.g., the fact that a particular patient received a particular dose of a particular fluid agent at a specified date, time and/or location and/or the current vaccination status of a patient (e.g., as a result of having participated in the vaccination event)), in accordance with a distributed chain code execution process as described herein. In accordance with some embodiments, the node devices **102a-n** may each store an instance of an Injection Management App as described herein, via which users of the node devices **102a-n** may communicate with an Injection Management System, which may own, control or operate (or have operated on its behalf) the server device **110** and an AI agent for conducting a pre-injection interrogation.

[0033] In some embodiments, the node devices **102a-n** may interface with the server device **110**

and/or the management device **106** to effectuate communications (direct or indirect) with one or more other node devices **102a-n** (such communication not explicitly shown in FIG. 1) operated by other users, for example. In some embodiments, the node devices **102a-n** may interface with the server device **110** to effectuate communications (direct or indirect) with the management device **106** (such communication also not explicitly shown in FIG. 1). In some embodiments, the node devices **102a-n** and/or the server device **110** may execute separate instances of a chain code algorithm that cause an injection event transaction ledger to be distributed in an encrypted and verifiable manner. As described herein, for example, the node devices **102a-n** and/or the server device **110** may communicate with the management device **106** to execute a cryptographic service utilized to securely disseminate an injection event chain code block or payload to a plurality of the node devices **102a-n**.

[0034] The network **104** may, according to some embodiments, comprise a Local Area Network (LAN; wireless and/or wired), cellular telephone, Bluetooth®, Near Field Communication (NFC), and/or Radio Frequency (RF) network with communication links between the server device **110**, the node devices **102a-n**, the management device **106**, and/or the memory devices **140-1a-n**, **140-2**. In some embodiments, the network **104** may comprise direct communications links between any or all of the components **102a-n**, **106**, **110**, **140-1a-n**, **140-2** of the system **100**. The node devices **102a-n** may, for example, be directly interfaced or connected to one or more of the server device **110** and/or the management device **106** via one or more wires, cables, wireless links, and/or other network components, such network components (e.g., communication links) comprising portions of the network **104**. In some embodiments, the network **104** may comprise one or many other links or network components other than those depicted in FIG. 1. The node devices **102a-n** may, for example, be connected to the server device **110** and/or the management device **106** via various cell towers, routers, repeaters, ports, switches, and/or other network components that comprise the Internet and/or a cellular telephone (and/or Public Switched Telephone Network (PSTN)) network, and which comprise portions of the network **104**.

[0035] While the network **104** is depicted in FIG. 1 as a single object, the network **104** may comprise any number, type, and/or configuration of networks that is or becomes known or practicable. According to some embodiments, the network **104** may comprise a conglomeration of different sub-networks and/or network components interconnected, directly or indirectly, by the components **102a-n**, **106**, **110**, **140-1a-n**, **140-2** of the system **100**. The network **104** may comprise one or more cellular telephone networks with communication links between the node devices **102a-n** and the server device **110**, for example, and/or may comprise the Internet, with communication links between the server device **110** and the management device **106** and/or one or more of the memory devices **140-1a-n**, **140-2**, for example.

[0036] The management device **106**, in some embodiments, may comprise any type or configuration of a computerized processing device, such as a PC, laptop computer, computer server, database system, and/or other electronic device, devices, or any combination thereof. In some embodiments, the management device **106** may be owned and/or operated by a third-party (i.e., an entity different than any entity owning and/or operating either the node devices **102a-n** or the server device **110**; such as certificate, authentication, and/or cryptographic service provider). The management device **106** may, for example, execute one or more web services that provide for centralized blockchain cryptographic functionality, such as the Hyperledger™ Fabric™ blockchain framework available from The Linux Foundation® of San Francisco, CA. In some embodiments, the management device **106** may receive block chain data from one or more of the node devices **102a-n** and/or the server device **110**, may apply a hash algorithm to the received data, and may transmit the encrypted data to each of the node devices **102a-n** and the server device **110** (e.g., for storage in local copies of a blockchain ledger). According to some embodiments, the management device **106** may comprise a plurality of devices and/or may be associated with a plurality of third-party entities.

[0037] In some embodiments, the server device **110** may comprise an electronic and/or computerized controller device, such as a computer server communicatively coupled to interface with the node devices **102a-n** and/or the management device **106** (directly and/or indirectly). The server device **110** may, for example, comprise one or more PowerEdge™ R830 rack servers manufactured by Dell®, Inc. of Round Rock, TX which may include one or more Twelve-Core Intel® Xeon® E5-4640 v4 electronic processing devices. In some embodiments, the server device **110** may comprise a plurality of processing devices specially-programmed to execute and/or conduct processes that are not practicable without the aid of the server device **110**. The server device **110** may, for example, execute one or more coded rules to manage a blockchain ledger for a plurality of injection event transactions, such management permitting real-time injection event status updates and adjustments which would not be capable of being conducted without the benefit of the specially-programmed server device **110**. According to some embodiments, the server device may comprise an Injection Management System as described herein, operable to communicate with patients and injection providers via their respective mobile devices (e.g., node devices **102a-n**) in order to obtain, update and transmit verifications of data stored in injection event records embodied in the injection event transaction ledger, injection event database or other repository of electronic records indicating injection events being tracked and managed by the Injection Management System and conduct a pre-injection interrogation via an AI agent.

[0038] According to some embodiments, the server device **110** may be located remote from one or more of the node devices **102a-n** and/or the management device **106**. The server device **110** may also or alternatively comprise a plurality of electronic processing devices located at one or more various sites and/or locations.

[0039] According to some embodiments, the server device **110** may store and/or execute specially programmed instructions to operate in accordance with embodiments described herein. The server device **110** may, for example, execute one or more programs, modules, and/or routines that facilitate the management, tracking, authorization, verification and/or updating of injection events and facilitating pre-injection interrogations using an AI agent, e.g., in an online environment, via an Injection Management App as described herein. According to some embodiments, the server device **110** may comprise a computerized processing device, such as a computer server and/or other electronic device to manage and/or facilitate transactions and/or communications regarding the node devices **102a-n**. A private company, government health agency, healthcare company, injector manufacturer and/or other user may, for example, utilize the server device **110** to (i) receive or recognize injection events that are associated with Unique ID injectors, (ii) conduct pre-injection interrogations, (ii) authorize injections from such injectors (e.g., based at least partially on answers provided during pre-injection interrogations), (iii) receive confirmations from injection providers or patients that a dose of a particular fluid agent has been administered to a particular patient (e.g., at a particular date/time and/or location); and (iv) store post-injection data that an AI agent may utilize to refine and/or modify questions comprising pre-injection interrogations, such as the questions to ask in particular sets of circumstances.

[0040] In some embodiments, the node devices **102a-n**, the management device **106**, and/or the server device **110** may be in communication with the memory devices **140-1a-n**, **140-2**. The memory devices **140-1a-n**, **140-2** may comprise, for example, various databases and/or data storage mediums that may store, for example, injector data, fluid agent data, registered patient data, registered injection provider data, injection event data, and/or injection or vaccination status data obtained from the node devices **102a-n**, digital receipt data (including privacy levels associated with digital receipts, in embodiments employing such levels) defined by the server device **110**, injection authorization processing rules, chain code instructions, blockchain data, cryptographic keys and/or data, login and/or identity credentials of registered users, and/or instructions that cause various devices (e.g., the server device **110**, the management device **106**, and/or the node devices **102a-n**) to operate in accordance with embodiments described herein.

[0041] The memory devices **140-1a-n**, **140-2** may store, for example, blockchain data defining a distributed injection event transaction ledger that stores records of injection event transactions (e.g., data defining injection events in which an injection has been authorized and/or administered to a particular patient and an injection provider has provided data indicating that a particular fluid agent has been administered to a particular patient, including attendant details such as the time/date, location, fluid agent and/or injection provider corresponding to the injection), chain code instructions, data that causes communications with the management device **106** (e.g., an API and/or API tunnel to a web service that provides blockchain authentication, certification, and/or cryptographic hashing). In some embodiments, the memory devices **140-1a-n**, **140-2** may comprise any type, configuration, and/or quantity of data storage devices that are or become known or practicable. The memory devices **140-1a-n**, **140-2** may, for example, comprise an array of optical and/or solid-state hard drives configured to store injection event transaction ledger data provided by (and/or requested by) the node devices **102a-n**, injector and/or injection event analysis data (e.g., analysis formulas and/or mathematical models), and/or various operating instructions, drivers, etc. While the memory devices **140-1a-n**, **140-2** are depicted as stand-alone components of the various node devices **102a-n** and the server **110**, the memory devices **140-1a-n**, **140-2** may comprise multiple components. In some embodiments, a multi-component memory devices **140-1a-n**, **140-2** may be distributed across various devices and/or may comprise remotely dispersed components. Any or all of the node devices **102a-n**, the management device **106**, and/or the server device **110** may comprise the memory devices **140-1a-n**, **140-2** or a portion thereof, for example.

[0042] Turning now to FIG. 2, illustrated therein is a block diagram of an Injection Management System **200** that may be a component of an Injection Management Service as described herein. The Injection Management System **200** may comprise, in one embodiment, an example of server device **110** (FIG. 1). The Injection Management System **200** may comprise, for example, a processor **201**, a memory **203**, a database **202** and a plurality of software modules **222-226**.

[0043] In accordance with some embodiments, any or all of the components of the Injection Management System **200** may comprise one or more hardware components, such as microprocessors, microcontrollers, or digital sequential logic, etc., such as processor **201**. The processor **201** may comprise one or more processors, such as one or more INTEL™ processors, working sequentially or in parallel. The processor **201** is in communication with, or operatively connected to, a memory **203**. The memory **203** may comprise an appropriate combination of magnetic, optical, and/or semiconductor memory, and may include, for example, Random Access Memory (RAM), Read-Only Memory (ROM), a compact disc and/or a hard disk. The processor **201** and the memory **203** may each be, for example: (i) located entirely within a single computer or other device; or (ii) connected to each other by a remote communication medium, such as a serial port cable, telephone line or radio frequency transceiver. In one embodiment, the system **200** may comprise one or more devices that are connected to a remote server computer for maintaining databases or injection event transaction records (e.g., a management device **106** of FIG. 1).

[0044] The system **200** may further comprise a database **202**, which may in some embodiments store data useful for implementing one or more embodiments described herein, non-limiting examples of which include (i) data associated with one or more users (e.g., patients receiving injections and/or injection providers), (ii) data associated with one or more drug companies whose drugs are packaged in injectors; (iii) data associated with injectors comprising data storage mechanisms storing unique identifiers; (iv) data associated with one or more injection events or injection event transactions; and (v) data associated with one or more rewards earned (or in the process of being earned) by patients on an injection compliance program, as described herein.

[0045] In accordance with some embodiments, the database **202** includes data, associated data structures, and database management software. The database **202** may, for example, be implemented using any well-known database management systems, including Microsoft SQL, Oracle, IBM DB2, etc. It should be noted that in some embodiments, database **202** (or at least some

of the data described as being stored therein) may be stored in memory **203** and/or in another memory device accessible to the memory **203** and/or to processor **201**. For example, in one embodiment database **202** (or at least some of the data described as being stored therein) may be stored in a memory of a third-party server, such as a server of a cloud-based computing service with which the Injection Management Service may contract for purposes of storing data.

[0046] In some embodiments, the data described herein as being stored in database **202** may be stored across more than one database; the example data described herein as being useful in at least some embodiments is described as being stored in a single database **202** merely for purposes of simplicity. In accordance with some embodiments, one or more of the types of data **204-212** may be stored as a separate database.

[0047] An example of a type of data that may be stored in database **202** includes injector data **204** defining a uniquely identifiable injector (and, in the case of pre-filled injectors, the dose of fluid agent packaged therein) recognized by the system (e.g., injectors which are registered with the system for purposes of being tracked, managed and/or authorized). Such injector data may include at least one of (i) a unique serial number of the injector (which may, in some embodiments, comprise a unique serial number of an NFC or other chip attached to or otherwise associated with the injector); (ii) a batch number for the fluid agent contained in the injector; (iii) a manufacturer's lot identifier of the fluid agent; (iv) a time and/or place of manufacture of the fluid agent and/or the injector; (v) the type, category and/or name of the fluid agent packaged in the injector; (vi) an expiration date of the fluid agent, if any; (vii) a strip ID number (e.g., if injectors are manufactured and distributed in strips of multiple injectors); (viii) a dosage or strength of the fluid agent; (ix) an intended destination or recipient (e.g., child vs. adult); (x) one or more characteristics of the injector (e.g., it is a 1.5 ml BFS vial); and/or (xi) other information desirable for identifying or tracking the Unique ID Injectors and/or the fluid agent contained within it (e.g., recall alerts). In some embodiments, the unique serial number may be a unique identifier that points to other data (e.g., at least some of the information indicated in items (ii)-(x) of the preceding list). In some embodiments, the injector data **204** may be utilized as part of an injection authorization process (e.g., to confirm that the injector is not counterfeit, the dose in it has not been subject to a recall, etc.).

[0048] Another example of a type of data that may be stored in database **202** includes patient data **206** defining patients who have registered with the Injection Management System **200** (e.g., individuals who have downloaded an Injection Management App in order to track their injection events, such as to track the vaccines they have received and their vaccination status for various infectious diseases), in accordance with at least some embodiments described herein. Such patient data **206** may include, in each respective patient record of the patient data **206**, at least one of: (i) a name; (ii) contact data; (iii) a unique identifier (e.g., as assigned by the Injection Management System or a third party organization that has partnered with the Injection Management System to help manage injection event information for individuals of that organization); (iv) login credentials (e.g., username and password via which the patient may log into a patient portal of the Injection Management App); (v) health history; and (vi) injection event status or vaccination status for various infectious diseases that the patient has been vaccinated for (or not). The health history may, for example, be updated periodically, in real time and/or responsive to certain events. For example, in some embodiments the health history may be updated based on answers provided in response to a pre-injection interrogation.

[0049] Yet another example of a type of data that may be stored in database **202** includes injection provider data **208** defining individuals who are registered with Injection Management System **200** as injection providers. Such injection provider data may include, for each record of a patient provider, at least one of: (i) a name; (ii) contact data; (iii) a unique identifier (e.g., as assigned by the Injection Management System or a third party that has partnered with the Injection Management System to assist with administering injections to patients via injectors registered with

the Injection Management System); and (iv) login credentials (e.g., username and password via which the patient may log into an injection provider portal of the Injection Management App). In some circumstances, a given individual may be registered with the Injection Management System as both a patient and an injection provider, in which case the individual would need to log in to the appropriate portal or platform (patient vs. injection provider) depending on the reasons for logging in.

[0050] Yet another example of the data that may be stored in database **202** is injection event data **210** defining a list of injection events or injection event transactions that have been recognized and recorded by the Injection Management System **200**. The injection event database may, in some embodiments, comprise an embodiment of a injection event transaction ledger as described with respect to FIG. **1**, or represent some or all of the same data as would be stored in the ledger. In accordance with some embodiments, each record of the injection event data **210** may store information defining a particular injection event that was recognized and recorded (e.g., in accordance with a method such as method **400** described with respect to FIG. **4**). Examples of the data that may be recorded in injection event data **210** include: (i) a date/time of an injection (it should be noted that the terms “date”, “time” and “date/time” are used interchangeably herein to refer to a date and/or time of an event); (ii) the location at which an injection was administered (e.g., based on information provided by an injection provider and/or patient, whether affirmatively by selecting from a list of choices or passively based on GPS or geolocation information obtained from a mobile device of either or both); (iii) an indication or identifier of the injection provider who administered the injection; (iv) an identifier of the patient to whom the injection was administered; (v) an indication of the injection (e.g., an indication of the dose and/or fluid agent that was administered, drug name, batch or lot number, etc.); (vi) an expiration date for the injection, if any; (vii) a unique identifier of an injector utilized in the injection event; (viii) a unique identifier for the transaction (if different from the injector identifier), which may be generated by the Injection Management System **200** upon a new record of injection event data **210** being opened; (ix) a passcode generated once the administration of the injection has been confirmed, which may expire; (x) an expiration time for the passcode and/or an indication of whether the passcode was received from the patient prior to the expiration time; (xi) a status of the injection corresponding to the injection event (e.g., if the injection is for a vaccine that is effective for a predetermined period of time, the stats may indicate whether the injection is currently considered valid/active vs. expired); and (x) a digital receipt or digital receipt identifier corresponding to the injection event.

[0051] Yet another example of the data that may be stored in database **202** is fluid agent data **212**. Fluid agent data **212** may comprise data regarding various fluid agents that are packaged in injectors being managed by Injection Management System **200**. In some embodiments, an injection provider may be able to look up information about a particular fluid agent pulled from fluid agent data **212** when administering the fluid agent to a patient. In other embodiments, the fluid agent data **212** may be utilized as part of an injection authorization process. Examples of information about a fluid agent that may be stored in fluid agent data **212** include, without limitation: (i) the ailment or disease that the fluid treats/prevents or a name for the fluid agent; (ii) a manufacturer of the fluid agent; (iii) ingredients included in the fluid agent (e.g., for allergy or compatibility purposes); (iv) distributor or contract manufacturer that produced the fluid agent; (v) contact information for a pharmaceutical company or other representative who may be able to provide additional information about the fluid agent, or for submitting concerns, complaints, questions or problems corresponding to the fluid agent; and (vi) drug interaction data, side effects, or conditions under which the fluid agent should not be administered (“drug effectiveness data”). In accordance with some embodiments, drug effectiveness data may be updated periodically, in real time and/or in response to certain events (e.g., upon receiving updated data from the manufacturer or distributor). In some embodiments, the AI agent facilitating a pre-injection interrogation may access and utilize the drug effectiveness data in formulating questions, processing answers and determining whether to

authorize an injection of a fluid agent (also referred to as a drug product herein).

[0052] In accordance with some embodiments, Injection Management System **200** may comprise a system for an artificial intelligence (AI) platform programmed to administer a pre-injection interrogation process to a patient prior to authorizing an injection from an RFID-enabled injector, by leveraging machine learning modeling and large data sets to generate and update one or more patient questions comprising the interrogation process. In some embodiments, such a system may comprise a server computing device communicably coupled to a database computing device and having a machine learning processor programmed with an instruction to execute an artificial intelligence algorithm. For example, the server computing device may be programmed to: (i) execute a pre-injection interrogation module (not shown) to generate at least one question comprising a pre-injection interrogation of a patient about to receive an injection of a drug product via an RFID-enabled injector containing the drug product, the at least one question designed to identify an increase likelihood of the patient experiencing at least one adverse side effect as a result of the injection; (ii) execute an injection side effect module (not shown) to generate, and maintain an updated version of, a drug product database. In one embodiment, each record of such a database may comprise: (i) a set of side effects corresponding to the drug product; and (ii) at least one patient characteristic that, if present at the time of injection, increases a likelihood of adverse side effects.

[0053] In some embodiments, the server computing device may be further programmed to (i) receive data comprising at least one input from at least one prior patient who has received at least one injection of the drug product, the at least one input indicating an experience of an adverse side-effect; (ii) update, based on the data, a record of the drug product database corresponding to the drug product; (iii) update, based on the update to the record, the pre-injection interrogation for the drug product as executed by the pre-injection interrogation module; and (iv) update a graphical user interface for display of the pre-injection interrogation via a display device, the graphical user interface including one or more visual representations of the one or more questions (e.g., a graphical user interface of a Injection Management App).

[0054] In some embodiments, the machine learning processor and the modules execute the artificial intelligence algorithm to decrease a likelihood of a patient experiencing adverse side effects by automatically assimilating newly-collected input data elements into the computer-generated models without relying on manual intervention. In some embodiments, the server computing device is further configured to build and update, by the machine learning processor, the computer-generated pre-injection interrogation module by training a machine learning algorithm programmed on the machine learning processor against a training data set

[0055] In some embodiments, the server computing device may be further programmed to: (i) determine, based on at least one answer provided in response to a question comprising the pre-injection interrogation process, that a corresponding patient is at an increased risk of at the least one adverse side effect; and (ii) applying at least one mitigation measure to mitigate the increased risk. For example, an RFID-enabled injector as described herein may be equipped with an output device and wherein applying the at least one mitigation measure comprises outputting an indication of an instruction comprising the at least one mitigation measure. Or, in another embodiment, the output device comprises a light and applying the at least one mitigation measure comprises lighting the light in a color indicating that the injection is not authorized.

[0056] In some embodiments, such a system may be further operable to update the graphical user interface to indicate the at least one mitigation measure, and apply the at least one mitigation measure comprises by outputting an instruction to the user of the RFID-enabled injector via the graphical user interface. For example, the instruction may comprise an instruction to contact a healthcare provider to discuss the increased risk of the at least one adverse side effect prior to proceeding with the injection or an instruction to decrease a dosage of the drug product. In some embodiments, such a server computing device may be further programmed to: (i) output, via the

graphical user interface at a time subsequent to the user receiving the injection via the RFID-enabled injector, a request for information describing at least one adverse side effect experienced by the user as a result of the injection of the drug product; and (ii) transmit any response to the request for information to the an injection side effect module.

[0057] It should be noted that the examples provided herein of what type of information may be included in which type of example data should not be taken in a limiting fashion. Modifications can be made as to what type of information is stored with which type of data, different types of data may be combined, some information may be stored with more than one type of data, etc.

[0058] The Injection Management System **200** may further comprise one or more software module(s) or engines for directing the processor **201** to perform certain functions. In accordance with some embodiments, software components, applications, routines or sub-routines, or sets of instructions for causing one or more processors to perform certain functions may be referred to as “modules” or “engines” herein. It should be noted that such modules, engine or any other software or computer program referred to herein (including the Injection Management App that can be downloaded to user's mobile devices for facilitating communications with the Injection Management Service, such as via the Injection Management Server **200**), may be written in any computer language and may be a portion of a monolithic code base, or may be developed in more discrete code portions, such as is typical in object-oriented computer languages. In addition, the modules, engines or any software or computer program referred to herein, may in some embodiments be distributed across a plurality of computer platforms, servers, terminals, and the like. For example, a given module may be implemented such that the described functions are performed by separate processors and/or computing hardware platforms.

[0059] With reference to FIG. **2**, it should be understood that any of the software module(s) or computer programs illustrated therein may be part of a single program or integrated into various programs for controlling processor **201**. Further, any of the software module(s) or computer programs illustrated therein may be stored in a compressed, uncompiled, and/or encrypted format and include instructions which, when performed by the processor **201**, cause the processor **201** to operate in accordance with at least some of the methods described herein. Of course, additional and/or different software module(s) or computer programs may be included and it should be understood that the example software module(s) illustrated and described with respect to FIG. **2** are not necessary in any embodiments. Use of the term “module” or “engine” is not intended to imply that the functionality described with reference thereto is embodied as a stand-alone or independently functioning program or application. While in some embodiments, functionality described with respect to a particular module may be independently functioning, in other embodiments such functionality is described with reference to a particular module for ease or convenience of description only and such functionality may in fact be a part of integrated into another module, program, application, or set of instructions for directing a processor of a computing device.

[0060] According to an embodiment, the instructions of any or all of the software module(s), engines or programs described with respect to FIG. **2** may be read into a main memory from another computer-readable medium, such from a ROM to RAM. Execution of sequences of the instructions in the software module(s), engines or programs causes processor **201** to perform at least some of the process steps described herein. In alternate embodiments, hard-wired circuitry may be used in place of, or in combination with, software instructions for implementation of the processes of the embodiments described herein. Thus, the embodiments described herein are not limited to any specific combination of hardware and software. Some non-limiting examples of software module(s) that may be utilized in system **200** include: (i) an injection management engine **220**; (ii) an authorization module **222**; (iii) an injection event transaction log module **224**; and (iv) an injection status module **226**.

[0061] In the example embodiment illustrated in FIG. **2**, the injection management engine is

illustrated as communicating with (i) an authorization module **222**; (ii) an injection event transaction log module **224**; (iii) an injection status module **226**; and (iv) the database **202**. An AI agent for facilitating a pre-injection interrogation may, in some embodiments, be a module of the injection management engine **220** or a module with which it interacts (e.g., it may comprise the authentication module **222**, since the answers to questions output during a pre-injection interrogation may determine whether an injection is to be authorized upon conclusion of the interrogation). Thus, the injection management engine is operable to access the data in the database **202** and, either itself or by providing such data to one or more of the modules **222-226**, perform some of the functionalities and embodiments described herein. Such an arrangement is illustrated as one example of how the data in database **202** may be accessed, modified and utilized.

[0062] Generally, the injection management engine **220** and the modules **222-226** should be understood as being accessible to processor **201**, irrespective of how in particular they are arranged within the system **200**, to implement one or more embodiments described herein. As described, one or more of the engine **220** and the modules **222-226** may be operable to utilize at least some of the data stored in database **202**. Further, in accordance with some embodiments, one or more of the engine **220** and the modules **222-226** may be operable to retrieve, manipulate, select, update, modify and/or determine data that is transmitted to and stored in database **202**.

[0063] Injection management engine **220** may, in accordance with some embodiments, operate to manipulate the data from database **202** in order to perform or facilitate one of the following example functionalities: (i) authorize an injection from a uniquely identifiable injector (e.g., based on information received from a patient during a pre-injection interrogation); (ii) store a record of each injection or dose of a fluid agent that has been administered by an injection provider utilizing the Injection Management System **200**; (iii) generate a digital receipt or other indication of an injection having been administered to a particular patient, to be output to the patient via the patient's mobile device (e.g., a node device **102a-n** of FIG. **1**); and/or (iv) allow a patient who has received an injection or dose of a fluid agent managed by Injection Management System **200** to provide verification of having received the injection or dose in an efficient, electronic, secure and verifiable manner.

[0064] In accordance with one embodiment, the injection management engine **220** or another component of system **200** may be operable to (i) output information to a user such as a patient or injection provider registered with the system **200** (such as questions comprising a pre-injection interrogation; and (ii) receive input or data from such users (such as an injector identifier, and responses or information responsive to a pre-injection interrogation), via one or more graphical user interfaces (GUIs) of an Injection Management App as described herein. An example of a process via which information may be received from, or output to, users such as injection providers and patients, is described herein with reference to FIG. **3**.

[0065] The interfaces of an Injection Management App may, for example, take the form of a Web server that conveys data in HTML, XML, or other well-known formats using well-known transmission protocols, such as HTTP and TCP/IP. Proprietary protocols and data formats may also be used.

[0066] Turning now to FIG. **3**, illustrated therein is an example process **300** which provides one embodiment of some functionality of an Injection Management System as described herein. Process **300** may, for example, be performed by Injection Management System **200** (FIG. **2**). In some embodiments, the process **300** may be performed and/or implemented by and/or otherwise associated with one or more specialized and/or specially-programmed computers (e.g., the processor **201** of FIG. **2**). It should be noted, with respect to process **300** and all other processes described herein, that not all steps described with respect to the process are necessary in all embodiments, that the steps may be performed in a different order in some embodiments and that additional or substitute steps may be utilized in some embodiments.

[0067] Turning now to FIG. **3**, process **300** is initiated at block **302** upon an initiation of an

injection event transaction being recognized by the system (e.g., Injection Management System **200**), which comprises receiving an injector identifier. In some embodiments, a patient identifier or other patient check-in process may precede block **302** (e.g., the patient may login to the App before the injection event begins in substance).

[0068] An initiation of an injection event transaction may be recognized, for example, based on a unique injector identifier being received from an injection provider via the Injection Management App. As described herein, each injector managed by an Injection Management Service may be equipped with, or correspond to, a data storage mechanism such as an NFC chip, RFID chip, QR code, bar code or other mechanism via which a sensor of a mobile device on which the Injection Management App is installed, and such data storage mechanism may store a unique identifier that uniquely identifies the injector. For example, the patient may be prompted to hold the injector close to the injection provider's mobile device or by tapping the injector (or a specific part of the injector, such as a label area in which the NFC chip is located) to the mobile device. In embodiments in which the data storage mechanism comprises something other than an NFC chip, the user may be prompted in a different manner to input the unique injector identifier, as appropriate. For example, the user may be prompted to scan a QR code or bar code on the injector or hold the injector comprising an RFID chip in close proximity to an RFID antenna that is operable to read the information from the RFID chip (which antenna may, in turn, be in communication with the injection provider's mobile device and/or Injection Management System in order to transmit the unique identifier of the injector and allow the injection event transaction to proceed).

[0069] Once the injection event transaction is initiated and the unique injector identifier is received in block **304**, the injector identifier is verified (e.g., as being authentic). For example, information associated with the injector identifier may be retrieved and an authorization process may be performed (e.g., to confirm that the injector is not expired, previously used, subject to a recall and/or otherwise not authorized for use). In accordance with some embodiments, the functionality described with respect to block **304** may be carried out at least partly by authorization module **222** (FIG. 2).

[0070] In one embodiment, once the injector identifier is received in block **304**, it is used to access data associated with the injector (as described with respect to FIG. 2). For example, a record of injector data **204** may be accessed by the Injection Management System **200** and one or more fields may be reviewed by a program or module (e.g., authorization module **222**) to confirm that there are no existing conditions that would cause the injector to not be authorized for use by the injection provider (e.g., that the injector is not subject to a recall; that the injector is authentic, which it may be presumed to be if the injector identifier is found in the injector data). In some embodiments, the Injection Management System may output information about the injector and/or fluid agent packaged in the injector to the patient or other injection provider via a GUI (not shown), as part of the injector authorization process. For example, the patient or other injection provider may be prompted to confirm that certain information printed or embossed on the injector or a label thereof matches information as stored in a database in association with the injector identifier.

[0071] In accordance with some embodiments, additional information or considerations may be involved in verifying an injector identifier and whether it the injector should be authorized for use (i.e., block **304** may comprise additional steps and/or authorization module **222** may perform additional verifications). For example, in some embodiments, a BFS vial or other type of injector may include a vaccine vial monitor (VVM) which may comprise an environmental monitoring mechanism (e.g., a strip adhered to the injector). In accordance with some embodiments, a VVM comprises a thermochromic label put on vials containing vaccines or other fluid agents which gives a visual indication of whether the vaccine or other fluid agent has been maintained within a temperature range which preserves its potency. The labels were designed in response to the problem of delivering vaccines to developing countries where environmental characteristics (e.g., the heat or cold within which the vaccines are stored) were difficult to maintain and track and

where formerly vaccines were being rendered inactive and administered ineffectively due to having been denatured by exposure to inappropriate temperatures (e.g., temperatures that are too cold for the vaccines). In some embodiments, the VVM may comprise a measure of a cumulative amount of heat or cold that the vial to which it is attached was exposed since it left the manufacturing facility. The label may turn black if the vial is exposed to an unacceptable temperature for an unacceptable period of time, a black label indicating that the vaccine or other fluid agent within the vial is no longer good and thus should not be injected. In other embodiments, a more expensive digital VVM may be utilized to track similar ambient temperature information.

[0072] In accordance with some embodiments in which an injector may include VVMs, an NFC chip or other component of the injector may comprise an optical chip monitor which allows for reading the output of a VVM (e.g., whether the VVM label has turned black). For example, a user may open the appropriate Injection Management App on their mobile device and tap the vial comprising the NFC chip and/or optical chip monitor to their mobile device (hold their mobile device (or camera of their mobile device) over the VVM) such that the app can read an indication of whether the vaccine is good and may be used, based on the VVM output. In another example, the user may open the appropriate Injection Management App on their mobile device and take a photo of the VVM output and the App may be programmed (or a server comprising a processor operable to implement an appropriate program, with which the App is in communication, may be so programmed) to determine, based on the output of the VVM, whether the fluid agent in the vial is good for use. In either example, if the Injection Management App determines based on the VVM output that the fluid agent is good to use, it may output to the user an indication that the user is authorized to inject the fluid agent (e.g., a green button, light or other indicator may appear in a screen of the Injection Management App), as described above with respect to otherwise authorizing an injector for use and providing an indication of such authorization to the injection provider. In one example, Applicant envisions a simple optical monitor included with/in the vial and operable to communicate with, or monitor the status of, the VVM such that if the VVM label (or indicator of a digital VVM) turns black (or otherwise indicates the fluid agent has been exposed to unacceptable temperatures), that causes a signal or determination that the fluid agent is no longer viable or authorized for use. The optical monitor may then be operable to communicate this indication to the NFC chip and/or the Injection Management App, and causes the NFC chip and/or the Injection Management App to determine that the fluid agent in the vial should not be authorized for injection into a patient.

[0073] Once the injector is authenticated by verifying the injector identifier as authentic and determining that there is no other information known to the Injection Management System that would cause the injector to not be authorized for use, the process proceeds to step **306**, in which the patient is uniquely identified. This may be done, for example, based on the login credentials that the patient provides when first logging into the Injection Management App in order to verify that the injector is authorized for use. In some embodiments, the patient may be prompted to further provide information or data to confirm his/her identity (e.g., a dual authentication process may be utilized, or the patient may be prompted to provide additional personal information, such as a portion of a social security number or driver's license). In some embodiments, facial recognition software (e.g., using a camera on the patient's smartphone) may be utilized to help identify the patient or verify the identity of the patient.

[0074] Applicant has recognized that the Injection Management App and processes enabled and described herein allow the Injection Management Service to obtain both data uniquely associated with the injector (and thus the fluid agent and perhaps even the particular dose of the fluid agent), as well as data uniquely associated with the patient (e.g., the patient's particular medical files or health history), at a point of injection. This set of unique data may be particularly useful for the benefit of the patient in cases where the patient is self-administering the injection and could significantly benefit from an AI agent utilizing this set of unique data, along with responses to a

pre-injection interrogation, to verify that the dose should in fact be self-administered or injected at that point in time (e.g., that there are no recalls or problems with the dose, no new known drug interactions or other drug effectiveness data relevant to the patient and that the patient is not experiencing any current medical condition (e.g., physical symptoms such as a fever) or has any updated medical history (e.g., is taking additional medicines that may have a newly identified drug interaction with the fluid agent in the injector identified in step **304**).

[0075] Once both the injector (and, according to some embodiments, the fluid agent and/or particular dose contained therein) are uniquely identified, along with the patient, in step **308** a set of customized questions are generated by the AI agent and output to the patient (directly or indirectly, such as via a healthcare professional or guardian). The questions may be determined or selected on information accessed based on the injector identifier. For example, the Injector Data **204**, the Fluid Agent Data **212** and/or the Patient Data **206** (all of FIG. 2) may be accessed by the Injection management Engine **220** (or Authorization Module **222**) to determine whether the patient should be asked questions pertaining to any particular drug interactions based on other medicines the patient is currently taking, risks associated with the patient's medical history or new information available regarding the fluid agent. As another example, the patient may be asked about his/her current physical symptoms or health based on a recent illness or illnesses known to be prevalent in the patient's geographical location.

[0076] In accordance with some embodiments, step **306** may also include providing an opportunity or mechanic for the patient to provide input, report problems or concerns and ask questions of the AI agent. For example, the patient may report having felt dizzy or nauseous after a prior injection of the same fluid agent. The AI agent may process this information to (i) determine additional questions to ask the patient; (ii) refine future questions to ask other patients going forward; (iii) update the patient's medical record (e.g., the Patient Data **206**) and/or determine whether to authorize the current injection of the fluid agent. For example, in analyzing data coming in over time for injections of the fluid agent contained in the injector identified in step **302**, the AI agent may have concluded that patients who experience feelings of dizziness and/or nausea after a first injection of the fluid agent are X % more likely to experience more severe or serious side effects after subsequent injections of the fluid agent and should thus not continue self-injecting the fluid agent without first consulting their physician. Thus, as patients report feeling dizzy and/or nauseous after receiving a first dose of a particular fluid agent and then subsequently report having been hospitalized or experiencing other more severe side effects from subsequent doses of the fluid agent, and this data is recorded in records accessible by the AI agent operable to facilitate the pre-injection interrogations as described herein, the AI agent may extrapolate from this that asking about feelings of dizziness after prior injection of the fluid agent is important and update its pre-injection interrogation questions for the particular fluid agent accordingly. The AI agent may also update its protocols to not authorize a self-injection of an additional dose of fluid agent for patients who received an earlier dose of the fluid agent and respond that they did feel dizzy afterwards, unless the patient also answers a question in the affirmative that they have since consulted their physician to report the feeling of dizziness and been told that they should continue to self-inject the fluid agent. Thus, answers to questions provided by patients during a pre-injection interrogation, as well as reporting of problems or concerns by the patient, may be utilized by the AI agent to not only update records, statistics and/or conclusions from data, but also to update, modify and/or refine the questions asked to patients.

[0077] In accordance with some embodiments, step **306** may also allow the patient to ask questions to the AI agent. For example, the patient may ask questions such as “is it OK to take this injection if I have a headache” or “I suspect I am developing a sinus infection, is it still OK to take this injection” or “is it OK to take this injection even though I am still recovering from a recent cold and am feeling achy”? To the extent the AI agent is able to provide answers to these questions based on its protocols, it can do so. To the extent the AI agent is not able to provide answers, it can

cause the patient to be put in contact with a live healthcare professional. For example, if the patient is utilizing the Injection Management App during process **300**, the AI agent may contact a healthcare professional in real time to join the pre-injection interrogation to chat with the patient and answer the question(s) and/or respond to the concerns the patient may have. In such embodiments, the healthcare professional may be tasked with providing input to the AI agent after interacting/chatting with the patient (e.g., to either proceed with authorizing the injection (or continue the pre-injection interrogation if it is not yet concluded) or that the injection should not be authorized).

[0078] During the pre-injection interrogation, as the customized questions generated in step **308** are output to the patient, the answers received from the patient are received (step **310**) and recorded (step **312**). As described herein, in some embodiments as the AI agent receives answers, an answer may cause the AI agent to formulate, retrieve or select a specific question based on that answer (or take some additional action, such as connecting the patient to a live healthcare provider). As also described herein, in some embodiments the answers may be recorded for future analysis and learning by the AI agent, so that the AI agent may modify or refine the pre-injection interrogation for future patients based on the recorded answers. Any complaints, concerns or input received from the patient during the pre-injection interrogation may likewise be recorded for subsequent use by the AI agent in learning and refining future pre-injection interrogation questions. For example, if a patient complains that they wish they had known that people prone to headaches are more likely to experience a bad headache as a side-effect of the fluid agent (e.g., because the patient is prone to headaches and did experience a bad headache after the last injection of the fluid agent), the AI agent may utilize this input to add to the pre-injection interrogation a question such as “Are you prone to frequent headaches?” and if the patient answers in the affirmative, outputting a warning such as “You may be more likely to experience a bad headache as a side effect of this injection than a patient who is not prone to frequent headaches.”

[0079] Once the patient's answers to the pre-injection interrogation are received and recorded, and any questions or concerns the patient may have are addressed, the Injection Management Engine **220** may determine whether the injection should be authorized (e.g., based on a prior verification of the injector or dose of fluid agent in the injector and on the answers to the pre-injection interrogation). If the injection is not to be authorized (e.g., the patient provided an answer about his/her current medical condition that leads the Injection Management Engine to conclude that the patient should not receive the injection until it is authorized by the patient's physician), the indication of this may be output to the patient via the Injection Management App (e.g., a “DO NOT INJECT” or “INJECTION NOT AUTHORIZED” message may be output, a red “X” may be output or another indication of the lack of authorization of the injection may be output). In embodiments in which the injector is equipped with a means of indicating lack of authorization (e.g., the injector comprises an LED light that can light up red if the injection is not authorized), the App may send a signal to the injector to output the appropriate indication to the patient.

[0080] In some embodiments, the AI agent may determine that one or more mitigation measures are to be implemented as a result of the interrogation process (e.g., as a result of one or more answers provided during the interrogation process). For example, the AI agent may determine that (i) a lower dosage of the drug product in the injector should be administered to the patient, to lessen the potential of adverse side effects; (ii) the injection should not be authorized until a mitigating condition is satisfied (e.g., the patient is put in touch with a healthcare professional who will discuss the increased likelihood of adverse side effects for the patient given one or more answers the patient provided during the interrogation process, and the healthcare provider then provides a verification to the AI agent that this discussion with the patient has taken place); and/or (iii) the patient should be informed to redo the interrogation process at a later time, in an attempt to provide answers that do not trigger additional mitigation measures. For example, if the patient currently has an increased heartrate, elevated blood pressure, cold, elevated temperature or headache, and one or

more of these conditions correspond to an increased likelihood of adverse side effects, the patient may be instructed to redo the interrogation process on another day when the patient is not experiencing the relevant condition(s)).

[0081] Assuming the injection is considered registered or verified, an indication of the authorized or verified injection is output to the patient (step **314**). This may comprise, for example, generating a record for the injection event transaction (e.g., opening a new record for the current injection event transaction in a database of the Injection Management System **200** (e.g., in injection event data **210** of FIG. **2**) and/or generating a unique transaction identifier for the injection event transaction). Further, once the injector is authenticated and authorized for use, a visual indication of this may be output to the patient or injection provider via a GUI Injection Management App (e.g., a checkmark may be output next to certain data or certain data may be highlighted or become activated). In some embodiments, additional visual, audio and/or tactile indications may be output to the injection provider to indicate that the injector has been authorized for use and that the injection provider should proceed with administering the injection to the patient (e.g., a symbol or other visual indication (e.g., checkmark, green light or thumbs up or completed circle), sound or vibration may be output via a GUI of the Injection Management App).

[0082] If the injector identifier is not successfully verified or the Injection Management System **200** determines that the injection is not authorized, an error, warning or other message indicating a failure to authorize the injection may be output to the injection provider via a GUI of the Injection Management App. For example, the injection provider may receive an indication that a recall has been issued for the injector (in which case the injection provider may attempt to verify a different injector and discard the rejected injector) or that the read of the injector identifier was invalid (in which case the injection provider may try to tap the injector to the mobile device again, scan the QR or other code again, or otherwise re-try to input the injector identifier, such as manually typing in a human-readable code of the injector).

[0083] It should be noted that some or all of the functions described herein with respect to blocks **302-312** may be performed by authorization module **222** (FIG. **2**) and/or injection event transaction log module **224** (FIG. **2**). For example, the injection event transaction log module **224** may serve to update or transmit updates regarding an injection event transaction to another device (e.g., node devices **102a-n** and/or management device **106**) in order to enable that device to update its records for the injection event transaction while the authorization module **222** may serve to perform functions that allow such updates/data to be confirmed or verified.

[0084] In some embodiments, the system may identify the patient and add an indication of the patient identifier to the injection event transaction record for the transaction. In some embodiments, updating the injection event transaction record with the patient identifier may be done as part of the process of block **412**.

[0085] In some embodiments, the Injection Management System **200** may require additional information or data to be provided by in the injection provider during an injection event before considering the injection event transaction to be completed and verified. In one example embodiment, the system may prompt the injection provider to take a photo or video of the patient who is receiving the injection and upload it as part of the injection event confirmation process or prior to considering an injection event transaction successfully completed.

[0086] In accordance with some embodiments, an NFC or RFID chip may be read-only chip while in other embodiments it may further be a writable chip to which information may be added. In some embodiments, the NFC or RFID chip may be a passive chip such that it does not include its own power source and may be powered by a reader device (e.g., a user's mobile phone) or other mechanism that is outside the NFC chip.

[0087] In embodiments in which the NFC or RFID chip is a writable chip (and perhaps other embodiments), there may be a need for a power supply or source for the chip. In one embodiment, a power source or supply may be included in an injector. In one embodiment, the power source

may be embedded in a BFS vial (e.g., a capacitor may be embedded in the BFS vial as part of the extrusion process). Other examples of sources or supplies of power that may be used to provide power to a NFC or RFID chip include, without limitation: (i) power created by a user from squeezing or pressing the BFS vial in order to inject the fluid agent within the vial; (ii) power created by a user shaking the BFS vial; (iii) power from a light or solar source included on/in the BFS vial; (iv) by use of piezoelectric effect and/or materials; (v) power from an external light, either natural as in sunlight, or by holding the device under a bright light source (e.g., in a plastic vial embodiment, the plastic of the vial could be embedded with a solar conversion chemical or material, wherein having the vial in the light provides a very low microvolt trickle charge to an associated battery or capacitor). It should be understood that in many embodiments, the amount of power needed for many of the embodiments described herein (if power for the NFC or RFID chip is needed at all) may be very small (e.g., one or two microvolts or even a fraction of a microvolt). [0088] In some embodiments, the NFC or RFID chip may be equipped with an output mechanism (e.g., a light), which may be used to indicate information to a user. For example, in some embodiments the light may be operable to output one of two or more colors, each color indicating different information. For example, a red light may be output to the user if an injection from the injector is not authorized as a result of an interrogation process and a green light may be output to the user if an injection from the injector is authorized as a result of the interrogation. In some embodiments, a mitigation measure as described herein may comprise outputting the red light, meaning that the injection is not being authorized.

[0089] In accordance with some embodiments, an NFC or RFID chip may become faulty, run out of power, become damaged, or rendered at least partially inoperable such that information on it cannot be rewritten or such that additional information cannot be written to it or stored within it. In some embodiments, an NFC or RFID chip may have limited capabilities (e.g., it can only be written to or rewritten a maximum number of times or within a maximum period of time). Thus, it may be desirable in some embodiments for an NFC or RFID chip to have the capability of confirming that the information that a user is attempting to write to it, or rewrite on it, has been successfully written or processed. Any and all of the information that is communicated to/from an NFC or RFID chip may, in at least some embodiments, be encrypted or secured or include a secure element, such that it is protected from unauthorized access.

[0090] In accordance with some embodiments, an NFC or RFID chip may include or provide GPS or location tracking capabilities. For example, the NFC or RFID chip may output or communicate the GPS coordinates of its location (e.g., periodically, non-periodically, in response to being interrogated by a sensor or in response to another condition being satisfied). In some embodiments, the GPS or location tracking functionality may provide for the NFC or RFID chip to determine and/or store (or provide information to allow another device, such as a user's mobile device using a corresponding software app to determine and/or store) information such as a current or past location of the vial corresponding to the NFC or RFID chip (e.g., where it has been and how long it has been there, where it currently is).

[0091] In accordance with some embodiments, the functionality of each NFC or RFID chip may be tested during the manufacturing process after it is embedded into the BFS vial, and any vials that are determined to include NFC chips that are not functionality properly or within desirable parameters may be discarded or routed to a specialized handling process. Thus, in one embodiment, the manufacturing process may provide for (i) adding an NFC or RFID chip to the vial (e.g., embedding the NFC or RFID chip into the plastic in a plastic extrusion manufacturing process or attaching the NFC or RFID chip under the cap of a glass vial or to another portion of a glass vial); (ii) filling the vial with the fluid agent and sealing it; (iii) testing the NFC or RFID chip to determine whether it is functioning within acceptable parameters (e.g., testing whether the unique identifier stored on it can be read successfully); and (iv) performing one of (a) allowing the vial to proceed to a first manufacturing pathway (e.g., to be packaged and sold as an unique ID injector) if

the NFC or RFID chip is tested successfully or (b) routing the vial to a second manufacturing pathway (e.g., to be discarded or to be sold as a non-NFC chip vial) if the NFC chip is not tested successfully). In accordance with some embodiments, an inkjet or other printer may also print an indication of the NFC or RFID chip identifier of an NFC or RFID chip embedded in, or attached to, a vial or component of a vial. The indication may be in human readable form (e.g., an alphanumeric code) or in machine readable form (e.g., a bar code or QR code). The indication may be printed directly on the vial or component of the vial (e.g., a cap, in the case of a glass vial) or onto a label that is then put on the vial. Thus, in such embodiments, the manufacturing process described above may further provide for (v) reading the NFC or RFID identifier of the NFC or RFID chip embedded in the vial; and (vi) causing a printing mechanism to print the NFC or RFID identifier and place it on the vial (either directly or as a label that is put on the vial).

[0092] In accordance with some embodiments, use of an Unique ID injector by a user in conjunction with an Injection Management App provides for a user (e.g., a patient into whom the fluid agent of the vial is injected) to be rewarded for a qualifying use (self-injecting the vaccine/medicine in the vial once it is authorized by the app and during an appropriate window of time). In one embodiment, an end user (a user who will be self-injecting the vaccine/medicine in the injector) downloads an Injection Management App onto his smart phone as a pre-requisite for allowing the user to earn rewards (e.g., extra minutes for phone use, extra data for a data plan, monetary payments, etc.) based on qualifying use of the vial (in other embodiments the user need not have the app downloaded and an off-line method for verifying the use may be implemented). For example, the user can earn extra phone minutes by tapping the vial to the smart phone (after opening up the app) during a qualifying window of time (e.g., during a time at which the user is supposed to be self-inject the vaccine/medicine in the injector based on a treatment regimen stored in the app). If the user taps the injector during a qualifying window, the user earns extra phone minutes or another reward (in some embodiments the injection may also additionally need to be authorized by the app based on information such as a verification that the dose in the injector is not expired or the VVM status of the injector is acceptable, as described herein).

[0093] Thus, in accordance with some embodiments, the Injection Management Service can effectively assign a unique identifier to a specific dose of a fluid agent (e.g., a specific dose of a vaccine) at the point of manufacture/filling of the injector, in a manner that does not readily allow a modification of the relationship between the dose and the unique identifier of the NFC or RFID chip that is embedded in or attached to the injector in which the dose is stored. This allows many benefits, including a tracking of when/where a particular dose of a fluid agent is injected into a patient and allowing a injection provider to readily verify that the dose in the injector about to be injected is in fact what it purports to be (e.g., by using the Injection Management App to read the unique identifier of the NFC or RFID chip of the injector prior to injection and having the app verify/authorize the dose for use).

[0094] In some embodiments, an NFC or RFID chip may be operable to provide for periodic, non-periodic (e.g., triggered by certain events or on a random basis) or continuous GPS tracking history of the location of the chip/injector (e.g., stored on the chip (e.g., in the cloud, on a server with which the NFC or RFID chip and/or a mobile device on which the Injection Management App is downloaded is operable to communicate). Additionally, the chip may communicate with a VVM, store a photo or indication of a status of a VVM or what otherwise be operable to determine a status of a VVM. This VVM status data can be stored locally on the chip and/or in another location (e.g., in the cloud and/or on a server with which the NFC or RFID chip and/or a mobile device on which the Injection Management App is downloaded is operable to communicate). Thus, in accordance with some embodiments, the NFC or RFID chip can know, with respect to the injector and the does of fluid agent it contains, one or more of: (i) where it is now, (ii) where it's been; (iii) what temperatures it has been exposed to over its long journey; and (iv) how long it's been since the day it was made and how much time has elapsed since then.

[0095] In accordance with some embodiments, an Injection Management App (via the NFC chip) and/or the NFC or RFID chip itself may be operable to recognize, store and/or forward to a server or other device various data that may be provided by a user, whether a injection provider, patient or other user. For example, as described herein, a user who is injecting a fluid agent from an NFC or RFID chip-enabled injector into a patient or a user who self-injecting the fluid agent may tap the vial to his/her mobile device prior to performing the injection such that data from the NFC or RFID chip is read and processed by the App. For example, the App may verify (based on information about the patient known to the app) that (i) the injector contains the appropriate fluid agent; (ii) it is being administered during an appropriate time; and (iii) the fluid agent is still good to inject (e.g., the VVM does not indicate that it has been exposed to inappropriate temperatures, an expiration date associated with the injector does not indicate the fluid agent is expired, there are no recalls associated with the vial, the vial is not a stolen vial, etc.). In some embodiments, the user may be requested to take a picture of the vial and/or the injection site at the time of the injection (in some embodiments the user may be rewarded for providing this information). In some embodiments, the user may be requested to provide an indication of side effects or reactions to the fluid agent (e.g., by answering one or more questions and/or uploading a picture of the injection site or the relevant side effect) using the App (and in some embodiments the user may be rewarded for providing such information).

[0096] In accordance with some embodiments, an Injection Management App may be utilized to communicate or follow up with users (e.g., patients who received injections or guardians or family members thereof). For example, a text message or App reminder may be sent to the mobile device of a user, reminding the user to take another self-injection of a fluid agent in accordance with a regimen. In another example, a text message or App reminder may prompt the user provide an indication of any reaction or side effect to an injection.

[0097] It should be noted that although it has been described herein that compliant or authorized injections from a Unique ID Injectors may be tracked via an app on a user's mobile device, a mobile device is not required for any embodiments described herein. Alternate devices may be used to help track the user's injections (e.g., the user may be provided with a SIM card or other electronic card that is operable to either store an indication of the user's injections directly or store an identifier that points to a cloud-based database which stores an indication of the user's injections).

[0098] It should further be noted that although an AI agent has been described herein as applied to an injection management platform to help identify relevant information from a patient during a pre-injection interrogation using customized questions and that the questions may be refined by the AI agent over time based on answers and patient input or complaints, such an AI driven process of questions may be applied to other applications or industries as well. For example, in a home-repair or home-improvement context, or any context in which a user is seeking services or a means of solving a particular problem, a similar AI agent may be employed for generating customized questions to help identify a service provider or solution in a way that evolves the questions over time based on user input, such that the AI agent learns and improves what are important questions to ask. Thus, such an AI agent may be trained over time to ask questions that may not seem obvious or intuitive but that it learns are important to users or would have improved results for the users if they had been asked, and will thus lead to improved future results for future users. For example, an AI agent for assisting users to find a relevant expert in a particular field may not initially ask the user whether the expert needs to have a high quality rating from persons in a particular demographic or from at least one person on the user's contact list is important to the user, but may learn to ask this question over time as users who are unsatisfied with the experts recommended by the AI agent turn out not to be satisfactory (as reported by the users to the AI agent) and the feedback from the users indicates that a reason for this may be that the recommended experts, although having a high quality rating, did not have a high quality rating from persons in a particular

demographic or who were known to the user seeking the expert.

RULES OF INTERPRETATION

[0099] Numerous embodiments have been described, and are presented for illustrative purposes only. The described embodiments are not intended to be limiting in any sense. The invention is widely applicable to numerous embodiments, as is readily apparent from the disclosure herein. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, and it is to be understood that other embodiments may be utilized and that structural, logical, software, electrical and other changes may be made without departing from the scope of the present invention. Accordingly, those skilled in the art will recognize that the present invention may be practiced with various modifications and alterations. Although particular features of the present invention may be described with reference to one or more particular embodiments or figures that form a part of the present disclosure, and in which are shown, by way of illustration, specific embodiments of the invention, it should be understood that such features are not limited to usage in the one or more particular embodiments or figures with reference to which they are described. The present disclosure is thus neither a literal description of all embodiments of the invention nor a listing of features of the invention that must be present in all embodiments.

[0100] The terms “an embodiment”, “embodiment”, “embodiments”, “the embodiment”, “the embodiments”, “an embodiment”, “some embodiments”, “an example embodiment”, “at least one embodiment”, “one or more embodiments” and “one embodiment” mean “one or more (but not necessarily all) embodiments of the present invention(s)” unless expressly specified otherwise.

[0101] The terms “including”, “comprising” and variations thereof mean “including but not limited to”, unless expressly specified otherwise.

[0102] The term “consisting of” and variations thereof mean “including and limited to”, unless expressly specified otherwise.

[0103] The enumerated listing of items does not imply that any or all of the items are mutually exclusive. The enumerated listing of items does not imply that any or all of the items are collectively exhaustive of anything, unless expressly specified otherwise. The enumerated listing of items does not imply that the items are ordered in any manner according to the order in which they are enumerated.

[0104] The term “comprising at least one of” followed by a listing of items does not imply that a component or subcomponent from each item in the list is required. Rather, it means that one or more of the items listed may comprise the item specified. For example, if it is said “wherein A comprises at least one of: a, b and c” it is meant that (i) A may comprise a, (ii) A may comprise b, (iii) A may comprise c, (iv) A may comprise a and b, (v) A may comprise a and c, (vi) A may comprise b and c, or (vii) A may comprise a, b and c.

[0105] The terms “a”, “an” and “the” mean “one or more”, unless expressly specified otherwise.

[0106] The term “based on” means “based at least on”, unless expressly specified otherwise.

[0107] The methods described herein (regardless of whether they are referred to as methods, processes, algorithms, calculations, and the like) inherently include one or more steps.

[0108] Therefore, all references to a “step” or “steps” of such a method have antecedent basis in the mere recitation of the term ‘method’ or a like term. Accordingly, any reference in a claim to a ‘step’ or ‘steps’ of a method is deemed to have sufficient antecedent basis.

[0109] Headings of sections provided in this document and the title are for convenience only, and are not to be taken as limiting the disclosure in any way.

[0110] Devices that are in communication with each other need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices that are in communication with each other may communicate directly or indirectly through one or more intermediaries.

[0111] A description of an embodiment with several components in communication with each other does not imply that all such components are required, or that each of the disclosed components

must communicate with every other component. On the contrary a variety of optional components are described to illustrate the wide variety of possible embodiments of the present invention.

[0112] Further, although process steps, method steps, algorithms or the like may be described in a sequential order, such processes, methods and algorithms may be configured to work in alternate orders. In other words, any sequence or order of steps that may be described in this document does not, in and of itself, indicate a requirement that the steps be performed in that order. The steps of processes described herein may be performed in any order practical. Further, some steps may be performed simultaneously despite being described or implied as occurring non-simultaneously (e.g., because one step is described after the other step). Moreover, the illustration of a process by its depiction in a drawing does not imply that the illustrated process is exclusive of other variations and modifications thereto, does not imply that the illustrated process or any of its steps are necessary to the invention, and does not imply that the illustrated process is preferred.

[0113] It will be readily apparent that the various methods and algorithms described herein may be implemented by, e.g., appropriately programmed general purpose computers and computing devices. Typically a processor (e.g., a microprocessor or controller device) will receive instructions from a memory or like storage device, and execute those instructions, thereby performing a process defined by those instructions. Further, programs that implement such methods and algorithms may be stored and transmitted using a variety of known media.

[0114] When a single device or article is described herein, it will be readily apparent that more than one device/article (whether or not they cooperate) may be used in place of a single device/article. Similarly, where more than one device or article is described herein (whether or not they cooperate), it will be readily apparent that a single device/article may be used in place of the more than one device or article.

[0115] The functionality and/or the features of a device may be alternatively embodied by one or more other devices which are not explicitly described as having such functionality/features. Thus, other embodiments of the present invention need not include the device itself.

[0116] The term “computer-readable medium” as used herein refers to any medium that participates in providing data (e.g., instructions) that may be read by a computer, a processor or a like device. Such a medium may take many forms, including but not limited to, non-volatile media, volatile media, and transmission media. Non-volatile media include, for example, optical or magnetic disks and other persistent memory. Volatile media may include dynamic random access memory (DRAM), which typically constitutes the main memory. Transmission media may include coaxial cables, copper wire and fiber optics, including the wires or other pathways that comprise a system bus coupled to the processor. Transmission media may include or convey acoustic waves, light waves and electromagnetic emissions, such as those generated during radio frequency (RF) and infrared (IR) data communications. Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, a carrier wave as described hereinafter, or any other medium from which a computer can read.

[0117] Various forms of computer readable media may be involved in carrying sequences of instructions to a processor. For example, sequences of instruction (i) may be delivered from RAM to a processor, (ii) may be carried over a wireless transmission medium, and/or (iii) may be formatted according to numerous formats, standards or protocols, such as Transmission Control Protocol, Internet Protocol (TCP/IP), Wi-Fi, Bluetooth, TDMA, CDMA, and 3G.

[0118] Where databases are described, it will be understood by one of ordinary skill in the art that (i) alternative database structures to those described may be readily employed, and (ii) other memory structures besides databases may be readily employed. Any schematic illustrations and accompanying descriptions of any sample databases presented herein are illustrative arrangements

for stored representations of information. Any number of other arrangements may be employed besides those suggested by the tables shown. Similarly, any illustrated entries of the databases represent exemplary information only; those skilled in the art will understand that the number and content of the entries can be different from those illustrated herein. Further, despite any depiction of the databases as tables, other formats (including relational databases, object-based models and/or distributed databases) could be used to store and manipulate the data types described herein.

[0119] Likewise, object methods or behaviors of a database can be used to implement the processes of the present invention. In addition, the databases may, in a known manner, be stored locally or remotely from a device that accesses data in such a database.

[0120] For example, as an example alternative to a database structure for storing information, a hierarchical electronic file folder structure may be used. A program may then be used to access the appropriate information in an appropriate file folder in the hierarchy based on a file path named in the program.

[0121] Computers, processors, computing devices and like products are structures that can perform a wide variety of functions. Such products can be operable to perform a specified function by executing one or more programs, such as a program stored in a memory device of that product or in a memory device which that product accesses. Unless expressly specified otherwise, such a program need not be based on any particular algorithm, such as any particular algorithm that might be disclosed in the present application. It is well known to one of ordinary skill in the art that a specified function may be implemented via different algorithms, and any of a number of different algorithms would be a mere design choice for carrying out the specified function.

[0122] Therefore, with respect to a means or a step for performing a specified function in accordance with 35 U.S.C. § 112, paragraph 6, structure corresponding to a specified function includes any product programmed to perform the specified function. Such structure includes programmed products which perform the function, regardless of whether such product is programmed with (i) a disclosed algorithm for performing the function, (ii) an algorithm that is similar to a disclosed algorithm, or (iii) a different algorithm for performing the function.

[0123] While various embodiments have been described herein, it should be understood that the scope of the present invention is not limited to the particular embodiments explicitly described. Many other variations and embodiments would be understood by one of ordinary skill in the art upon reading the present description.

Claims

1. A system for an artificial intelligence (AI) platform programmed to administer a pre-injection interrogation process to a patient prior to authorizing an injection from an RFID-enabled injector, by leveraging machine learning modeling and large data sets to generate and update one or more patient questions comprising the interrogation process, the system comprising: an RFID-enabled injector comprising a single-dose vial containing a drug product and having attached thereto an RFID tag for enabling remote communication with a server computing device, the RFID tag comprising an external light element; and a server computing device communicably coupled to a database computing device and having a machine learning processor programmed with an instruction to execute an artificial intelligence algorithm, the server computing device programmed to: (a) receive data from the RFID-enabled injector, the data indicating that a patient is about to receive an injection of the drug product from the RFID-enabled injector; (b) execute a pre-injection interrogation module to generate at least one question comprising a pre-injection interrogation of the patient, the at least one question designed to identify an increase likelihood of the patient experiencing at least one adverse side effect as a result of the injection; (c) execute an injection side effect module to generate, and maintain an updated version of, a drug product database, each record of the database comprising (i) a set of side effects corresponding to the drug product; and (ii) at

least one patient characteristic that, if present at the time of injection, increases a likelihood of adverse side effects; (d) receive data comprising at least one input from at least one prior patient who has received at least one injection of the drug product, the at least one input indicating an experience of an adverse side-effect; (e) update, based on the data, a record of the drug product database corresponding to the drug product; (f) update, based on the update to the record, the pre-injection interrogation for the drug product as executed by the pre-injection interrogation module; and (g) update a graphical user interface for display of the pre-injection interrogation via a display device, the graphical user interface including one or more visual representations of the one or more questions, wherein the machine learning processor and the modules execute the artificial intelligence algorithm to decrease a likelihood of a patient experiencing adverse side effects by automatically assimilating newly-collected input data elements into the computer-generated models without relying on manual intervention, and wherein the server computing device is further configured to build and update, by the machine learning processor, the computer-generated pre-injection interrogation module by training a machine learning algorithm programmed on the machine learning processor against a training data set.

2. The system of claim 1, wherein the server computing device is further programmed to: determine, based on at least one answer provided in response to a question comprising the pre-injection interrogation process, that a corresponding patient is at an increased risk of at the least one adverse side effect; and applying at least one mitigation measure to mitigate the increased risk.
 3. The system of claim 1, wherein the RFID-enabled injector is equipped with an output device and wherein applying the at least one mitigation measure comprises outputting an indication of an instruction comprising the at least one mitigation measure.
 4. The system of claim 3, wherein the output device comprises a light and wherein applying the at least one mitigation measure comprises lighting the light in a color indicating that the injection is not authorized.
 5. The system of claim 1, wherein the system is operable further update the graphical user interface to indicate the at least one mitigation measure, and wherein applying the at least one mitigation measure comprises outputting an instruction to the user of the RFID-enabled injector via the graphical user interface.
 6. The system of claim 5, wherein the instruction comprises an instruction to contact a healthcare provider to discuss the increased risk of the at least one adverse side effect prior to proceeding with the injection.
 7. The system of claim 5, wherein the instruction comprises an instruction to decrease a dosage of the drug product.
 8. The system of claim 1, wherein the server computing device is further programmed to: output, via the graphical user interface at a time subsequent to the user receiving the injection via the RFID-enabled injector, a request for information describing at least one adverse side effect experienced by the user as a result of the injection of the drug product; and transmit any response to the request for information to the injection side effect module.
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